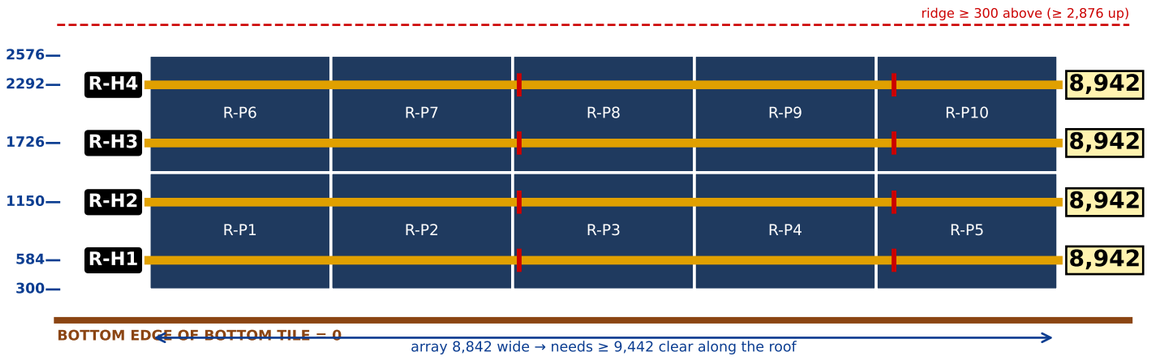
JAKEMAN – 15 Willow Drive, Bicester OX26 3XA | 22 PANELS LANDSCAPE | 9–10 Nov 2026 | PAGE 1/4 – LAYOUT & RAILS

v4 (John 9 Nov): RAILS HORIZONTAL – SSE gable flipped (lone panel TOP, valley side) – along the roof, parallel to the eaves. Panels LANDSCAPE per Regen proposal 11213122. 2 rails per row. Faces drawn as you STA

REAR – SW face (garden side) – 10 panels
az 241° (SW) | pitch 38° Regen / 33° survey phone



Regen proposal aerial (north up)

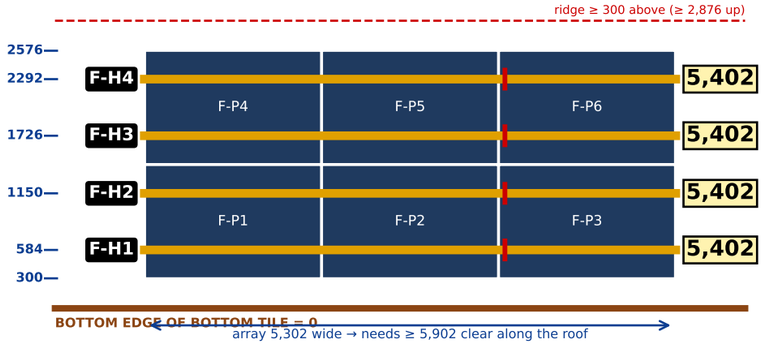


RAIL CUT LIST – K2 SingleRail 36 (full bars from LEFT, only END piece cut

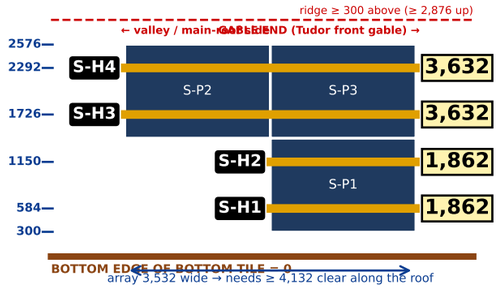
Rails	Panels	LENGTH	3.65 m bars	Joints ea	Height up (mm)
R-H1, R-H2	5	**8,942**	2 full + 1,642	2	584 / 1150
R-H3, R-H4	5	**8,942**	2 full + 1,642	2	1726 / 2292
F-H1, F-H2	3	**5,402**	1 full + 1,752	1	584 / 1150
F-H3, F-H4	3	**5,402**	1 full + 1,752	1	1726 / 2292
S-H1, S-H2	1	**1,862**	cut 1,862	0	584 / 1150
S-H3, S-H4	2	**3,632**	cut 3,632	0	1726 / 2292
N-H1, N-H2	1	**1,862**	cut 1,862	0	584 / 1150
N-H3, N-H4	2	**3,632**	cut 3,632	0	1726 / 2292

Yellow = K2 rail (label left, LENGTH right). Red tick = rail joint (splicer).
Blue = mm UP from the bottom tile to the rail centre.

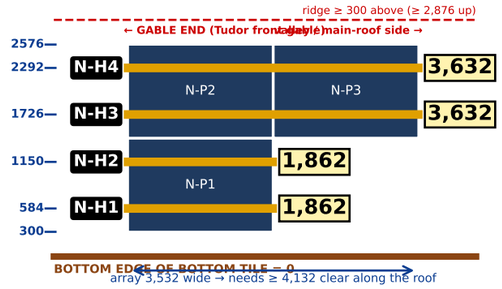
FRONT MAIN – NE face (drive side) – 6 panels
az 61° (NE) | pitch 38° Regen



FRONT GABLE – SSE face – 3 panels
az 152° (SSE) | pitch 49° Regen



FRONT GABLE – NNW face – 3 panels
az 332° (NNW) | pitch 44° Regen



Rail = n × 1762 + 8 × (n−1) + 100 (50 past the end panel each end – same K2 method as before).

16 rails: 4 × 8,942 • 4 × 5,402 • 4 × 3,632 • 4 × 1,862

BARS: 22 × 3.65 m • SPLICERS (rail connectors): 12

Clamps: 32 end (30 mm) + 28 mid (8 mm gap)

Hooks (ESTIMATE – rafters TO CONFIRM): ≈ 76 at 400 or 600 c/c (hooks ≤1200 apart); ≈ 104 at 450 c/c (≤900). Add one where a joint isn't between two hooks.

RAIL SPACING: 2 rails per row, centres 284 in from the top & bottom (long) edges of the panel = 566 apart – EJD/Regen short-side method (Regen/JA OK 19 Oct 26). JA manual has NO short-side zone for JAM54D41-LR; the nearest JA figure (other models) is 0–235 mm from the corner. TO CONFIRM WITH REGEN BEFORE FIXING.

Joints: nearest is 60 mm from a panel frame (K2 min 50) – keep the clamp off the splicer.

Horizontal rails remove v2's rafter-alignment risk (rails no longer have to sit on a rafter) – but EVERY HOOK must still land on a rafter.

Not for upload – John's info only. Not a structural design. Every roof dimension on this sheet is MEASURED ON SITE – nothing has been assumed.

All 'up' figures are FROM THE BOTTOM EDGE OF THE BOTTOM TILE (0). Across positions need roof widths measured on site – none were supplied.

A SET-OUT – UP THE ROOF (same heights on every face)

Mark	mm UP from bottom tile
Bottom panel row – bottom edge	**300**
H1 – 1st rail centre (row 1 lower)	**584**
H2 – row 1 upper rail	**1150**
Bottom row – top edge	**1434**
Mid gap between rows (8)	**1434–1442**
Top row – bottom edge	**1442**
H3 – row 2 lower rail	**1726**
H4 – row 2 upper rail	**2292**
Top row – top edge	**2576**
Rails along the roof: start 50 before the 1st panel, end 50 past the last	**—**
Ridge must be at least (on slope)	**2,876**

300 above the bottom tile = John's minimum edge rule. If John wants the arrays higher/centred, every figure moves up by the same amount.

ACROSS (along the eaves): REAR & FRONT – centre the block between the verges unless John says otherwise (check the TV aerial at the NW end of the ridge). GABLES – lone panel (own short pair of rails) sits on the GABLE-END side (as drawn), ≥300 from the gable verge and ≥300 clear of the valley. ALL ACROSS FIGURES TO CONFIRM ON SITE.

C STILL TO CONFIRM ON SITE

- ☐ **Roof width along the eaves on each face – rear needs ≥ 9,442, front ≥ 5,902, each gable face ≥ 4,132 at the 2-panel row (≥300 each end incl. valley).**
- ☐ **Bottom tile → ridge on the slope, each face – must be ≥ 2,876.**
- ☐ Pitch: Regen 38° main / 49° SSE gable / 44° NNW gable; survey phone 33° – check (two faces of one gable shouldn't differ).
- ☐ **Rafter c/c on every face → hook count & positions (hooks on rafters, ≤1200 apart, ~300 in from panel ends, one either side of each joint).**
- ☐ **RAIL SPACING 284 in / 566 apart (short-side clamping) – confirm with Regen/JA before fixing.**
- ☐ Gable faces: lone panel side and valley clearance as drawn (from Regen aerial – check against the roof).
- ☐ Tile make / profile and hook type delivered.
- ☐ Panel label = JA JAM54D41-460/LR, 1762 × 1134 × 30.
- ☐ Optimisers: NONE listed in Regen's quote – if Tigos turn up, ask Regen which panels before fitting.
- ☐ TV aerial / dish / vents near the rear or front arrays.

B BIRD MESH CUT LIST (Bird guard: YES) – total ≈ 66.6 m

Each piece = side length + 150 each end (wraps round both corners) = side + 300. Go clockwise from the top-left of each block.

REAR – SW face (garden side) (10 panels)

#	Where	Side	CUT	✓
1	Top (along roof)	8,842	**9,142**	
2	Right side (up roof)	2,276	**2,576**	
3	Bottom (along roof)	8,842	**9,142**	
4	Left side (up roof)	2,276	**2,576**	
	TOTAL	22,236	**23,436**	

FRONT MAIN – NE face (drive side) (6 panels)

#	Where	Side	CUT	✓
1	Top (along roof)	5,302	**5,602**	
2	Right side (up roof)	2,276	**2,576**	
3	Bottom (along roof)	5,302	**5,602**	
4	Left side (up roof)	2,276	**2,576**	
	TOTAL	15,156	**16,356**	

FRONT GABLE – SSE face (3 panels)

#	Where	Side	CUT	✓
1	Top – along both top panels	3,532	**3,832**	
2	Down the outer side of the 2-high column	2,276	**2,576**	
3	Bottom of the bottom panel	1,762	**2,062**	
4	Up the inside of the bottom panel	1,134	**1,434**	
5	Under the lone top panel	1,770	**2,070**	
6	Down the outer side of the lone top panel	1,134	**1,434**	
	TOTAL	11,608	**13,408**	

FRONT GABLE – NNW face (3 panels)

#	Where	Side	CUT	✓
1	Top – along both top panels	3,532	**3,832**	
2	Down the outer side of the 2-high column	2,276	**2,576**	
3	Bottom of the bottom panel	1,762	**2,062**	
4	Up the inside of the bottom panel	1,134	**1,434**	
5	Under the lone top panel	1,770	**2,070**	
6	Down the outer side of the lone top panel	1,134	**1,434**	
	TOTAL	11,608	**13,408**	

TOTAL CUT ≈ 66,608 mm = 66.6 m. Count the roll(s) BEFORE cutting – one 30 m roll is NOT enough.

Fit after panels are on; tape the real runs before cutting. Keep mesh clear of valleys / flashings.

Measure these BEFORE the first hook and send to John. Layout is from Regen's proposal – roof sizes were NOT supplied.

1 JOB

JAKEMAN – 15 Willow Drive, Bicester, OX26 3XA

Mon 9 & Tue 10 Nov 2026 • Installer: EJD

Regen: Dale Collins 07933 391 307 / 01327 577 811 • Quote 11213122 • SC no: _____

22 × JA Solar JAM54D41-460/LR (1762 × 1134 × 30) • ALL LANDSCAPE

Rear SW 10 (2 × 5) • Front NE 6 (2 × 3) • Front gable SSE 3 • Front gable NNW 3

Roof: concrete tiles • Scaffold: YES • Bird guard: YES

K2 SingleRail 36, RAILS HORIZONTAL (2 per row) + K2 hooks for concrete tile

Optimisers: none in Regen's quote. Inverter/battery = electrician, not us.

REAR – SW face: 10 panels (2 rows × 5)



FRONT – NE face 6 panels; Tudor gable (left) carries SSE 3 + NNW 3



Regen aerial



Compass



Pitch 33°



2 MEASURE & RECORD (write in, photo, send to John)

	REAR SW (10)	FRONT NE (6)	GABLE SSE (3)	GABLE NNW (3)
Width along eaves (mm)				
Needs at least	9,442	5,902	4,132	4,132
Bottom tile → ridge on slope (mm)				
Needs at least	2,876	2,876	2,876	2,876
Pitch (°)				
Rafter c/c (mm)				
1st rafter from LEFT verge (mm)				
Obstructions (what/where)				
Tile make / profile				
Start time / finish time				

Any measurement under its 'needs at least' → STOP, ring John before fixing anything on that face.

Record times: start ___ hooks done ___ rails done ___ panels done ___ mesh done ___ finish ___

Asbestos: 1990s house – not expected. Anything suspect → STOP, don't break it, ring John.

Do it in this order, one face at a time (rear first). Anything that doesn't match → STOP and ring John. NO electrical work – the electrician does all cabling/connections.

FITTING ORDER

SET OUT

1. Page 3 measured and sent to John; every face passes its 'needs at least'. Rail spacing confirmed with Regen.
2. Mark 0 = BOTTOM EDGE OF THE BOTTOM TILE COURSE. Every 'up' figure (page 2) is from here.
3. Mark the rail lines UP from 0: H1 584, H2 1150, H3 1726, H4 2292. Chalk them ALONG the roof, following the tile courses.
4. Mark the block ends along the eaves (centred / ≥300 from verges & valleys). Rails start 50 before the first panel.
5. Find every rafter (loft hatch on landing + lift tiles). Crayon rafter centres where they cross each rail line.

HOOKS – must be on rafters

6. On each rail line: end hooks on the rafter nearest ~300 in from the panel ends (250–350), then hooks on rafters no more than 1200 apart (400/600 c/c) or 900 (450 c/c).
7. A hook either side of every rail joint, joint ≥150 from a hook (joints marked red on page 1).
8. Lift the tile, screw the hook to the rafter centre (K2 screws per manual). Arm in the tile VALLEY, not on the roll. Notch the tile above if it lifts. Re-hang every tile flat.

RAILS – HORIZONTAL

9. **Rails run ALONG the roof. Labels page 1: R-H1–R-H4, F-H1–F-H4, S-H1–S-H4, N-H1–N-H4 (H1 = lowest). Marker label + length on every rail.**
10. Full 3.65 m bars from the LEFT, only the END piece cut. Join with the K2 splicer – never under a clamp.
11. Rails follow the TILE LINE (no spirit level / string line). Lay a spare rail UP the slope across all the rails – every rail must touch it; pack any hook that doesn't. Tighten T-bolts.

PANELS

12. Bottom row first, from the left. Landscape (long side along the roof). End clamps (30 mm) at each end of each row; mid clamps between panels (8 mm gap). Torque to K2 figures.
13. Gables: lone panel sits on its own short pair of rails on the gable-end side as drawn. Check valley clearance.

BIRD MESH

14. Cut per page 2 (≈67 m – unchanged). Clip round the outside of every block, tight to the tiles.

OPTIMISERS

15. None in Regen's quote. If Tigos arrive: don't fit until Regen say which panels; panel leads into Tigo IN, OUT leads left capped for the electrician.

FINISH

16. Every hook, T-bolt, splicer and clamp tight, tiles flat, waste off the roof. Photos of each face + times to John.

TOOLS – mounting, asbestos, working at height ONLY

MOUNTING

☐ Rip saw + spare blades ☐ File / deburr ☐ Tape ×2 ☐ Chalk line ☐ Marker / crayon

☐ Square ☐ Cordless driver + bits & sockets ☐ Impact driver

☐ **Torque wrench + K2 clamp / T-bolt torque settings**

☐ Grinder + stone disc / tile cutter ☐ Tile lifter ☐ Hook packers ☐ Spare rail (flatness)

☐ 2–3 spare concrete tiles ☐ WIRE CUTTERS + STANLEY KNIFE (mesh) ☐ Phone (photos)

WORKING AT HEIGHT

☐ Scaffold tag checked ☐ Roof ladder (gable faces) ☐ Harness + lanyard ☐ Hard hat, gloves, glasses, hi-vis, boots

☐ Rope / bucket for hoisting ☐ Roof protection mat

ASBESTOS (precaution)

☐ FFP3 masks ☐ Disposable coveralls ☐ Sealable bags + tape ☐ Water spray ☐ Wet wipes

☐ **Suspect material → STOP, don't drill/break, ring John.**

CHECK REGEN DELIVERY

Count on arrival (needed from the layout):

☐ 22 × JA JAM54D41-460/LR

☐ 22 × K2 rails 3.65 m

☐ 12 × rail splicers / connectors

☐ Hooks ≈ 76 (400/600 c/c) – ≈ 104 if 450 c/c

☐ Hook screws – per K2 manual

☐ 32 × end clamps 30 mm

☐ 28 × mid clamps

☐ T-bolts + nuts = 1 per hook

☐ Bird mesh ≥ 69 m + clips

☐ Tigos? none expected

Short or wrong → ring Regen BEFORE going up.

NUMBERS TO REMEMBER

Panel 1762 × 1134 × 30, LANDSCAPE

Rails HORIZONTAL, 2 per row

Rails 284 in from panel top & bottom (566 apart)

Rail = n×1762 + 8×(n−1) + 100

8,942 • 5,402 • 3,632 • 1,862

H1 584 • H2 1150 • H3 1726 • H4 2292 up

Hooks on rafters ≤1200 apart

Mid gap 8 • end clamp 30

Panels ≥300 from every edge

0 = bottom edge of bottom tile

Dale 07933 391 307

Not for upload – John's info only. Not a structural design. Every roof dimension on this sheet is MEASURED ON SITE – nothing has been assumed.